

Ascomylactams A–C, Cytotoxic 12- or 13-Membered-Ring Macrocyclic Alkaloids Isolated from the Mangrove Endophytic Fungus *Didymella* sp. CYSK-4, and Structure Revisions of Phomapyrrolidones A and C

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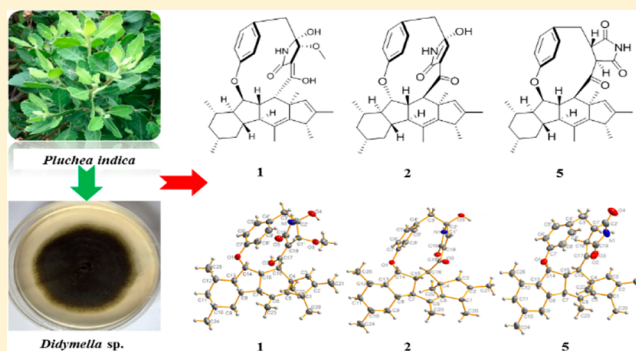
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S Supporting Information

ABSTRACT: Three new 12- or 13-membered-ring macrocyclic alkaloids, named ascomylactams A–C (1–3), along with the analogues phomapyrrolidone C (4) and phomapyrrolidone A (5) were isolated from the mangrove endophytic fungus *Didymella* sp. CYSK-4. Their structures were elucidated by analysis of extensive spectroscopic data and mass spectrometric data. The structures and absolute configurations of 1 and 2 were determined by single-crystal X-ray diffraction experiments, which represents the first crystal structures described for a (6/5/6/5) tetracyclic skeleton fused with a 12- or 13-membered-ring macrocyclic moiety. The configurations of phomapyrrolidone C (4) and phomapyrrolidone A (5) were revised by detailed analysis of the NMR data. In a cytotoxic assay, compounds 1 and 3 showed moderate cytotoxicity against MDA-MB-435, MDA-MB-231, SNB19, HCT116, NCI-H460, and PC-3 human cancer cell lines, with IC₅₀ values in the range of 4.2–7.8 μM.



Decahydrofluorene-class^{1,2} compounds have generated considerable attention for their highly complex 12- or 13-membered macrocycles and biological activities.^{3–5} For example, GKK1032s,⁶ pyrrocidines,^{7–10} and hirsutellones^{11,12} have a tricyclic decahydrofluorene skeleton and have shown antimicrobial, cytotoxic, apoptosis-inducing,¹³ and prolyl oligopeptidase inhibitory activities. Similarly the embellicines¹⁴ and phomapyrrolidones,¹⁵ with a cyclopenta[*b*]fluorene (6/5/6/5) ring system, showed antitubercular, cytotoxic, cytostatic, and NF-κB transcriptional inhibitory activities. However, compounds with a tetracyclic skeleton fused with a 12- or 13-membered macrocyclic moiety are relatively rare in the decahydrofluorene class.

Mangrove endophytic fungi have proven to be a rich source of structurally novel and biologically active secondary metabolites that have become interesting and significant resources for drug discovery.¹⁶ As part of our ongoing search for new antitumor active compounds from mangrove

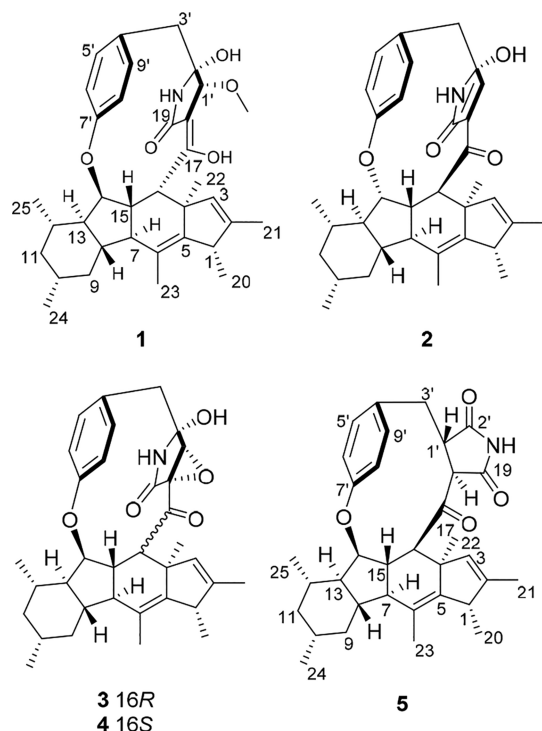
endophytic fungi from the south China sea,^{17–19} the strain *Didymella* sp. CYSK-4, isolated from a healthy branch of the marine semimangrove *Pluchea indica*, was investigated. We recently reported three dichloroisocoumarins from this fungus²⁰ and now report three new decahydrofluorene-class natural products, ascomylactams A–C (1–3), which are constructed of a 6/5/6/5 tetracyclic core modified with an additional 12- or 13-membered-ring macrocyclic moiety. The configurations of phomapyrrolidones A (5) and C (4) also have been revised. Their structures and absolute configuration were determined using single-crystal X-ray diffraction analysis and comparisons of experimental and calculated ECD spectra. Herein, the isolation, structure elucidation, structure revisions, and cytotoxic activities of these compounds are reported.

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RESULTS AND DISCUSSION

The south China sea fungus *Didymella* sp. CYSK-4 was cultured on solid rice medium with seawater for 28 days. The EtOAc extract of the fermentation media and mycelia was fractionated by repeated silica gel chromatography and Sephadex LH-20 column chromatography to afford the compounds 1–5.



Ascomylactam A (1) was obtained as white crystals. The molecular formula was determined as $C_{35}H_{45}NO_5$ based on the HRESIMS peak at 558.3222 $[M - H]^-$, indicating 14 degrees of unsaturation. The 1H NMR spectrum (Table 1) displayed four aromatic protons at δ_H 7.13 (d, $J = 8.5$ Hz, H-5'), 6.96 (m, H-6'), 6.96 (m, H-8'), and 7.04 (d, $J = 8.1$ Hz, H-9'), seven methyl groups at δ_H 1.09 (d, $J = 7.1$ Hz, H₃-20), 1.67 (s, H₃-21), 1.01 (s, H₃-22), 1.83 (s, H₃-23), 0.94 (d, $J = 6.5$ Hz, H₃-24), and 1.17 (d, $J = 6.2$ Hz, H₃-25), and one oxygenated methyl at δ_H 3.41 (s, 1'-OMe). There are 35 carbon signals in the ^{13}C NMR spectra (Table 1), indicating the presence of seven methyls (one oxygenated), three methylenes, 15 methines (five olefinic and two oxygenated), 10 nonprotonated carbons (one carbonyl and seven olefinic), and one oxygenated olefinic carbon (Table 1). Three spin systems from H-7 to H-16, including H₃-24 and H₃-25, from H-1 to H₃-20, and from H-2 to H₃-21 were deduced from COSY correlations (Figure 1). Along with the HMBC correlations (Figure 1) from the methyl group H₃-35 to C-1, C-2, C-5, and C-16, from H-2 to C-4 and C-34, from H₃-33 to C-3, C-4, and C-5, and from H₃-23 to C-5, C-6, and C-7, a 6/5/6/5 tetracyclic skeleton could be constructed for 1. The HMBC correlations from NH-20 to C-18, C-1', C-2', and C-19, from OH-2' to C-1' and C-2', and from H-1' to C-18, C-2', and C-19 implied the presence of a γ -lactam moiety. Further cross-peaks from H₂-3' to C-1', C-2', C-4', and C-5' indicated that a 1,4-disubstituted aromatic ring was linked to the γ -lactam moiety via a methylene (C-3'; δ_C 46.3). The correlations of H-14 with C-7', as well as the deshielded C-14 (δ_C 88.6) and C-7' (δ_C 160.5), indicated the

linkage of C-14 and C-7' via an oxygen atom. Then, the correlations of H-16 with C-17/C-18 and H-1' with C-17/C-18 further confirmed an unusual 13-membered macrocycle fused to the tetracyclic system. Further HMBC correlations from OH-17 to C-18, together with the chemical shift of C-17 (δ_C 172.1), indicated the attachment of a hydroxy group. Comprehensive analyses of the NMR data implied that compound 1 was similar to embellicine B,¹⁴ except for the appearance of a methoxy group OCH₃-1' (δ_H 3.41, δ_C 59.1) in compound 1, which was located at C-1' based on the HMBC correlation from 1'-OMe to C-19. Thus, the planar structure of compound 1 was established as shown.

The relative configuration of 1 was determined by analysis of the NOESY data (Figure 2). The key correlations of H-6'/H-14, H-14/H-7, H-7/H₃-22, H-7/H₃-20, H-6'/H₃-25, H₃-25/H-11b, H-11b/H₃-24, and H-11b/H-13 suggested that they were positioned on the same face, while the correlations of H-10/H-8, H-8/H-15, H-8'/H-16, H-16/H-1', and H-1'/H-9' revealed they were positioned on the other face. Furthermore, the key correlations supported by the NMR spectra were confirmed by the X-ray diffraction data (Figure 3). Thus, the relative configuration of compound 1 was defined as (1S*, 4R*, 7S*, 8S*, 10R*, 12S*, 13R*, -14R*, 15S*, 16S*, 1'R*, 2'R*). In addition, the correlations of NH-20/H-5', H-14/H-6', H-1'/H-9', H-16/H-8', H-1'/H-16, and H₃-25/H-6' indicated that the benzene ring had restricted rotation and appeared to have a parallel orientation to the γ -lactam moiety, which was further confirmed by the X-ray structure. As compound 1 shares a biogenetic pathway with compound 2 (vide infra), the absolute configuration of 1 was determined as 1S, 4R, 7S, 8S, 10R, 12S, 13R, 14R, 15S, 16S, 1'R, and 2'R. Incubation of compound 2 with MeOH under various conditions did not result in the formation of 1, supporting 1 as a natural product (see Experimental Section).

Ascomylactam B (2), isolated as a white solid, exhibited a negative HRESIMS ion peak at m/z 526.29608 $[M - H]^-$, indicating its molecular formula as $C_{34}H_{41}NO_4$. The NMR data (Table 1) of 2 were similar to those of 1, except for one methine at δ_C 158.6, one quaternary carbon at δ_C 129.2, and one carbonyl carbon at δ_C 197.6. Furthermore, the disappearance of the 1'-OMe (δ_C 59.1, δ_H 3.41) and an exchangeable signal OH-17 (δ_H 12.11) indicated the presence of a double bond between C-18–C-1' and a ketone carbonyl group at C-17 in 2. This deduction was supported by the HMBC correlations (Figure 1) from H-1' to C-17, C-18, C-2', and C-19, from H-16 to C-17, and from NH-20 to C-18, C-1', and C-2'. The relative configuration of 2 was established by analysis of the NOESY spectrum (Figure 2). The structure and absolute configuration were further confirmed by a crystallographic analysis. A block crystal was obtained from MeOH, and subsequently single-crystal X-ray diffraction analysis using Cu K α radiation gave the Flack parameter of 0.04(12) (Figure 3), which allowed assignment of the absolute configuration of 2 as 1S, 4R, 7S, 8S, 10R, 12S, 13R, 14S, 15S, 16R, and 2'R. Compound 2 (ascomylactam B) has the same planar structure as embellicine A, which was isolated from the fungus *Embellisia eureka*.¹⁴ The configurations at C-1, C-4, C-7, C-14, and C-16 in 2 were different from embellicine A. Numerous chemical shifts and NOE correlations differed between 2 and embellicine A in DMSO- d_6 . In addition, NMR shift differences between compound 2 and embellicine A (Table S12), specific rotations (Table S5), and an experimental ECD spectrum of 2 (Figure S50) are available in the Supporting Information.

Table 1. ^1H (500 MHz) and ^{13}C (125 MHz) NMR Data for Compounds 1–3

position	1		2		3	
	$\delta_{\text{C}}^{\text{a}}$ type	$\delta_{\text{H}}^{\text{a}}$ (mult, J (Hz))	$\delta_{\text{C}}^{\text{a}}$ type	$\delta_{\text{H}}^{\text{a}}$ (mult, J (Hz))	$\delta_{\text{C}}^{\text{b}}$ type	$\delta_{\text{H}}^{\text{b}}$ (mult, J (Hz))
1	44.8, CH	3.05, q (6.8)	43.8, CH	2.90, d (7.01)	45.9, CH	2.75, q (7.3)
2	140.4, C		143.1, C		144.5, C	
3	134.1, CH	5.43, s	130.9, CH	5.18, s	131.4, CH	5.31, s
4	52.8, C		53.5, C		56.1, C	
5	145.2, C		145.5, C		144.4, C	
6	129.4, C		136.3, C		129.9, C	
7	48.3, CH	2.09, d (9.5)	45.6, CH	2.29, t (11.3)	48.1, CH	2.09, t (11.5)
8	57.5, CH	1.95, m	44.7, CH	1.23, m	45.3, CH	1.62, m
9a	40.5, CH_2	2.14, d (12.2)	40.6, CH_2	2.16, d (12.4)	41.2, CH	2.18, d (12.6)
9b		0.72, overlap		0.90, dd (12.4, 24.2)		0.80, m
10	32.5, CH	1.54, m	32.9, CH	1.55, m	33.8, CH	1.55, m
11a	45.2, CH_2	1.79, m	45.2, CH_2	1.75, d (14.2)	46.6, CH_2	1.82, m
11b		0.72, overlap		0.75, dd (11.9, 24.4)		0.71, m
12	31.7, CH	1.88, m	36.6, CH	1.53, m	32.8, CH	1.86, m
13	56.7, CH	1.05, m	59.8, CH	1.32, m	58.0, CH	1.26, m
14	88.6, CH	4.69, dd (4.3, 7.9)	81.2, CH	4.52, t (7.1)	86.5, CH	4.82, m
15	43.1, CH	1.60, dd (2.8, 11.1)	51.0, CH	2.07, m	55.3, CH	2.0, dt (6.3, 12.5)
16	47.1, CH	1.95, m	53.2, CH	3.28, d (10.5)	58.4, CH	3.76, d (6.2)
17	172.1, C		197.6, C		203.9, C	
18	104.0, C		129.2, C		60.0, C	
19	173.7, C		169.3, C		170.8, C	
1'	80.4, CH	3.51, s	158.6, CH	6.80, s	67.7, CH	3.56, brs
2'	88.3, C		87.8, C		84.8, C	
3a'	46.3, CH_2	3.22, d (13.3)	45.5, CH_2	3.32, d (12.7)	45.1, CH_2	3.05, brs
3b'		2.94, d (13.3)		3.17, d (12.7)		
4'	128.8, C		126.6, C		129.1, C	
5'	130.6, CH	7.13, d (8.5)	130.9, CH	7.01, d (8.6)	132.4, CH	7.07, dd (2.0, 8.3)
6'	123.1, CH	6.96, m	122.3, CH	6.97, d (8.9)	119.8, CH	7.19, dd (2.3, 8.3)
7'	160.5, C		159.1, C		158.9, C	
8'	122.7, CH	6.96, m	114.3, CH	6.78, d (8.6)	122.9, CH	6.60, dd (2.3, 8.2)
9'	131.9, CH	7.04, d (8.1)	132.1, CH	6.96, d (8.9)	133.3, CH	7.10, dd (2.0, 8.2)
20	19.1, CH_3	1.09, d (7.1)	18.8, CH_3	1.03, d (7.1)	19.5, CH_3	0.98, d (4.5)
21	14.7, CH_3	1.67, s	14.8, CH_3	1.51, s	14.6, CH_3	1.53, s
22	23.9, CH_3	1.01, s	29.6, CH_3	1.43, s	27.2, CH_3	1.27, s
23	15.4, CH_3	1.83, s	15.6, CH_3	1.78, s	15.9, CH_3	1.80, s
24	22.7, CH_3	0.94, d (6.5)	22.6, CH_3	0.95, d (6.4)	22.9, CH_3	0.97, d (3.9)
25	20.3, CH_3	1.17, d (6.2)	20.3, CH_3	1.01, d (6.5)	20.6, CH_3	1.10, d (6.1)
1'-OMe	59.1, CH_3	3.41, s				
17-OH		12.11, s				
2'-OH		3.86, s				
20-NH		5.99, s		6.73, s		4.57, s

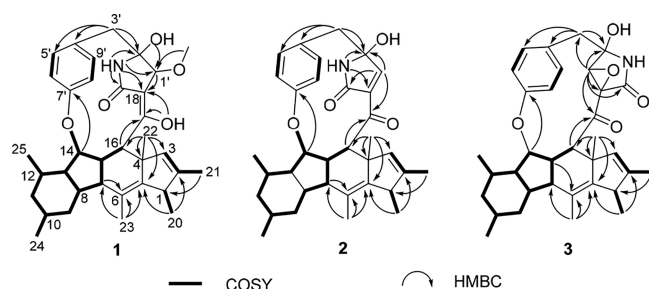
^aMeasured in CDCl_3 . ^bMeasured in $\text{MeOH}-d_4$.

Figure 1. Key HMBC and COSY correlations of compounds 1–3.

Thus, compound 2 (ascomylactam B) and embellicine A were isomeric.

Ascomylactam C (3) was obtained as a white solid, and its molecular formula was determined to be $\text{C}_{34}\text{H}_{41}\text{NO}_5$ by

HRESIMS and ^{13}C NMR data. The differences between 2 and 3 were that the oxygenated carbons C-18 (δ_{C} 60.0) and C-1' (δ_{C} 67.7) in 3 replaced the olefinic carbons C-18 (δ_{C} 129.2) and C-1' (δ_{C} 158.6) in 2, respectively. This indicated the presence of an epoxide in the γ -lactam moiety, which was further confirmed by the correlations (Figure 1) from H-1' to C-17, C-18, C-19, C-2', and C-3'. In the NOESY spectrum (Figure 2), the correlations of H-14/H-6', H-6'/H-16, H-16/H-7, H-7/H-13, H-16/H-22, H-22/H-20, H-25/H-6', H-25/Ha-11, and Ha-11/H-24 revealed the cofacial relationship of these protons, while the cross-peaks of H-15/H-8', H-8'/H-19, and H-15/H-8 showed these groups on the opposite side of the molecule. Moreover, the correlations of NH-20/H-5', H-14/H-6', H-25/H-6', H-1'/H-9', H-1'/H-8', and H-15/H-8' indicated that the benzene ring and γ -lactam moiety of 3 were nearly parallel oriented, the same as compound 1, and afford

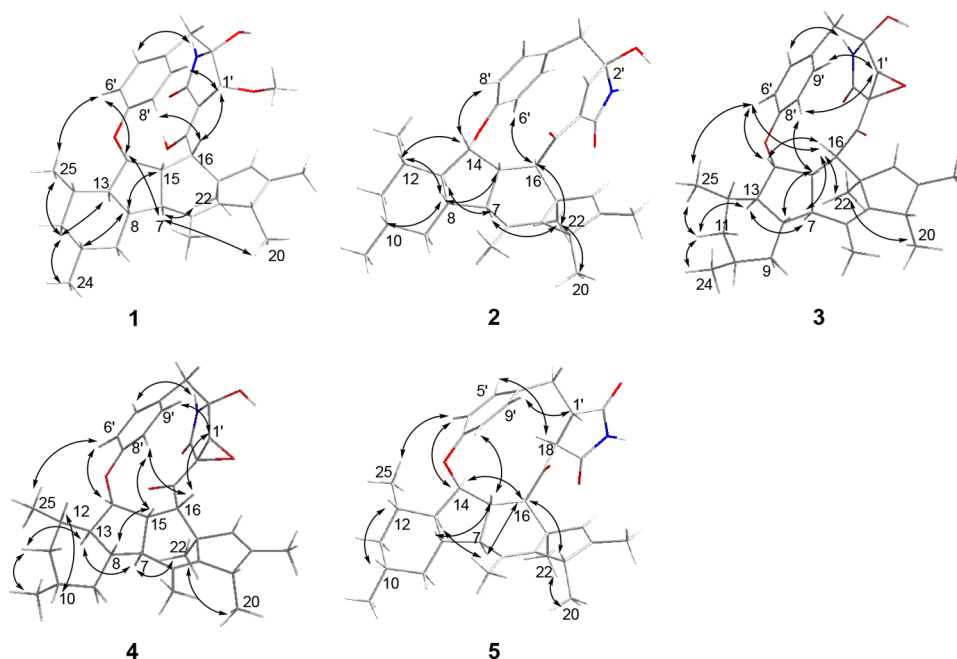


Figure 2. Key NOE correlations of compounds 1–5.

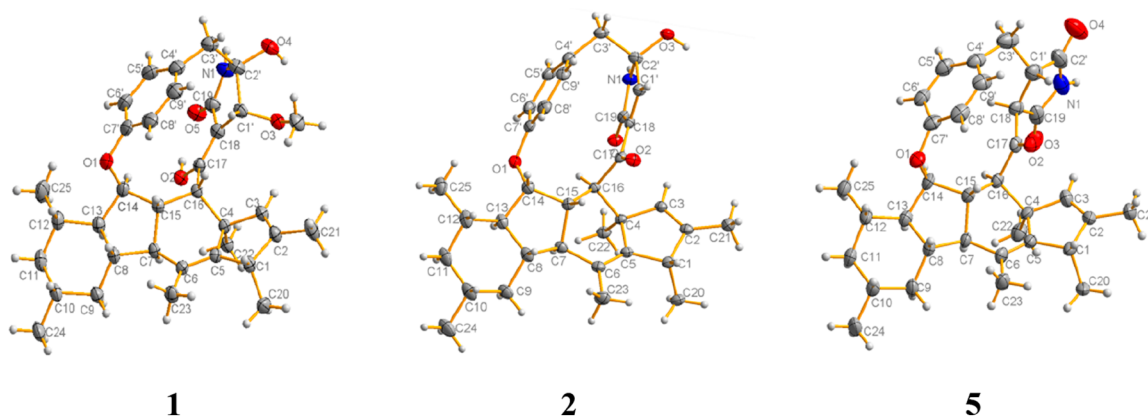


Figure 3. X-ray crystallographic analysis of compounds 1, 2, and 5.

the 2'*R* configuration in compound 3. Subsequently, no correlations from OH-2' to H-1' or H-9' were observed in DMSO-*d*₆ (Figure S25b), which further supported the C-2' configuration.

To define its absolute configuration, the DFT-calculated electronic circular dichroism (ECD) spectrum of 3 obtained at the B3LYP/3-21G level was compared with the experimental ECD spectrum. The experimental ECD spectrum of 3 exhibited negative Cotton effects at 225, 250, and 315 nm, as well as a weak positive Cotton effect at 275 nm, which was in a good agreement with that of the calculated one (Figure 4). Thus, the absolute configuration of 3 was assigned to be 1*S*, 4*R*, 7*S*, 8*S*, 10*R*, 12*S*, 13*R*, 14*R*, 15*S*, 16*R*, 18*R*, 1'*R*, and 2'*R*.

Compound 4, with a molecular formula of C₃₄H₄₁NO₅, the same as 3, was obtained as a white solid. The NMR data (Table S13) closely resembled those of 3, except for the chemical shifts in the signals at C-4 ($\Delta\delta_C$ -3.7), C-14 ($\Delta\delta_C$ +2.4), C-15 ($\Delta\delta_C$ +2.9), C-16 ($\Delta\delta_C$ -4.8), and C-22 ($\Delta\delta_C$ -1.7). The NOESY correlations (Figure 2) of H-16/H-1', H-16/H-8', and H-15/H-8' observed in compound 4 confirmed that 4 was an epimer of 3 at C-16. The absolute configurations

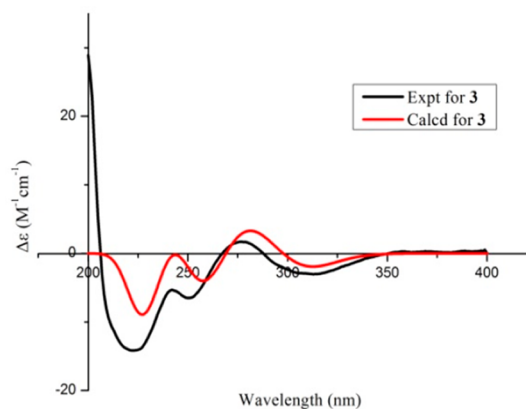


Figure 4. Experimental and calculated ECD spectra of compound 3 in MeOH.

were deduced to be 1*S*, 4*R*, 7*S*, 8*S*, 10*R*, 12*S*, 13*R*, 14*R*, 15*S*, 16*S*, 18*R*, 1'*R*, and 2'*R* by comparison of the theoretical ECD spectrum with the experimental one (Figure 5). Furthermore,

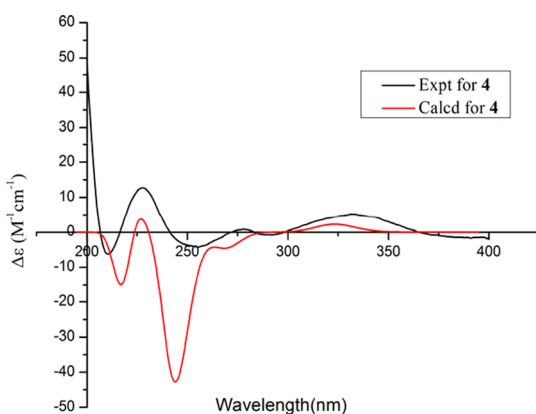


Figure 5. Experimental and calculated ECD spectra of compound 4 in MeOH.

comparison of the NMR data (Table S9) for compound 4 (measured in CDCl_3) with those of phomapyrrolidone C^{15,21} indicated that they are identical. But, the relative configurations at C-1, C-4, and C-7 in phomapyrrolidone C, which were determined by the correlations of H-7 to H-15, H-15 to H-16, and H-16 to H₃-22, were different from those of compound 4. With the consideration of biogenetic origin, the configurations at C-1, C-4, and C-7 in 4 were assigned to be 1S, 4R, and 7S, the same as those of 1 and 2, which also were supported by the correlations of H-7/H-13, H-7/H₃-22, and H₃-22/H₃-20 (Figure 2). Thereby, the relative configuration of phomapyrrolidone C has been revised and the absolute configuration has been assigned as 1S, 4R, 7S, 8S, 10R, 12S, 13R, 14R, 15S, 16S, 18R, 1'R, and 2'R. In addition, chemical shift assignments at C-4, C-6, C-5', C-6', C-8', and C-9' in phomapyrrolidone C also have been revised in the Supporting Information.

Compound 5 has a molecular formula of $\text{C}_{34}\text{H}_{41}\text{NO}_4$ shown by HRESIMS. The NMR data for compound 5 (Table S11) (measured in CDCl_3) were consistent with phomapyrrolidone A,²¹ suggesting that 5 was identical to phomapyrrolidone A. The relative configurations of 5 at C-1, C-4, C-7, C-18, and C-1' were determined by the correlations of H-7/H-16, H-16/H₃-22, H₃-22/H₃-20, H-16/14, H-14/8', H7-/H-13, H-18/H-5', and H-1'/H-9' (Figure 2), which were further confirmed by a single-crystal X-ray diffraction experiment using Cu K α radiation (Figure 3). The absolute configuration of 5 was assigned to be 1S, 4R, 7S, 8S, 10R, 12S, 13R, 14R, 15S, 16R, 18R, and 1'S by identical calculated and experimental ECD spectra (Figure 6). Because the relative configuration of 5 was firmly established by the X-ray analysis, the structure of phomapyrrolidone A has been revised to the 1S, 4R, 7S, 8S, 10R, 12S, 13R, 14R, 15S, 16R, 18R, and 1'S configuration. Chemical shift assignments at C-5', C-6', C-8', and C-9' in phomapyrrolidone A also have been revised in the Supporting Information.

The cytotoxic activities of compounds 1–5 against the MDA-MB-435 (melanoma), MDA-MB-231 (breast cancer), SNB19 (glioblastoma), HCT116 (colon cancer), NCI-H460 (lung carcinoma), and PC-3 (prostatic cancer) human cell lines were evaluated using the MTT method,²² with epirubicin as the positive control. Compounds 1 and 3 showed moderate cytotoxicity against all cell lines, with IC_{50} values ranging from 4.2 to 7.8 μM , and compound 2 exhibited cytotoxicity against the MDA-MB-231 and HCT116 cell lines with IC_{50} values of 6.6 and 4.5 μM , respectively (Table 2).

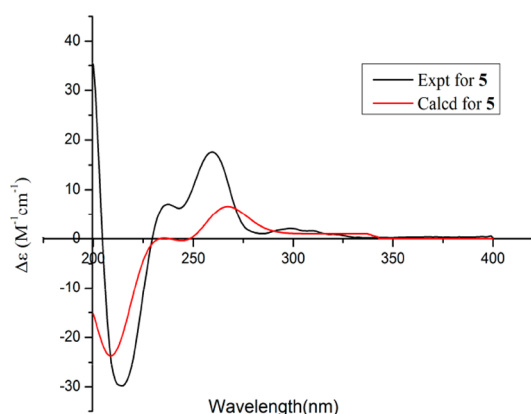


Figure 6. Experimental and calculated ECD spectra of compound 5 in MeOH.

Table 2. Cytotoxicity Data of Compounds 1–5 (IC_{50} , μM)

compound	MDA-MB-435 ^b	MDA-MB-231 ^c	SNB19 ^d	HCT116 ^e	NCI-H460 ^f	PC-3 ^g
1	4.9	5.9	6.8	5.5	4.4	5.7
2	12	6.6	18	4.5	13	20
3	7.8	5.1	7.8	4.2	4.4	7.5
4	16	12	29	15	12	22
5	26	30	>50	25	28	>50
EPI ^a	0.26	5.6	2.3	0.37	0.12	5.7

^aPositive control. ^bHuman melanoma cell line. ^cHuman breast cancer cell line. ^dHuman glioblastoma cell line. ^eHuman colon cancer cell line. ^fHuman lung carcinoma cell line. ^gHuman prostatic cancer cell line.

Compounds 1–5 are likely biosynthesized by a pathway similar to GKK1032A₂.²³ Related compounds from the decahydrofluorene-class family GKK1032s, pyrrocidines, and hirsutellones, embellicines, and phomapyrrolidones are known to have cytotoxic activity. Moreover, a patent has reported the antitumor activity of GKK1032s. Thus, the antitumor activities of compounds like the ascomylactams that contain a 12- or 13-membered macrocycle and a tetracyclic skeleton are worth studying further.

EXPERIMENTAL SECTION

General Experimental Procedures. Melting points were recorded on a Fisher-Johns hot-stage apparatus and were uncorrected. Optical rotations were measured on an MCP 300 (Anton Paar) polarimeter at 28 °C. UV spectra were recorded in MeOH using a PERSEE TU-1900 spectrophotometer, and ECD data were obtained on a Chirascan CD spectrometer (Applied Photophysics). IR spectra were recorded on a Nicolet Nexus 670 spectrophotometer, in KBr discs. All NMR experiments were performed on a Bruker Avance 500 spectrometer at room temperature. All chemical shifts (δ) were given in ppm with reference to tetramethylsilane. ESIMS spectra were measured on a Finnigan LCQ-DECA mass spectrometer, and HRESIMS spectra were obtained on a Thermo Fisher Scientific Q-TOF mass spectrometer. Column chromatography (CC) was conducted using silica gel (200–300 mesh, Qingdao Marine Chemical Factory) and Sephadex LH-20 (Amersham Pharmacia). Thin-layer chromatography (TLC) was performed on silica gel plates (Qingdao Huang Hai Chemical Group Co., G60, F-254).

Fungal Material. The fungus CYSK-4 used in this research was isolated from a healthy branch of the marine semimangrove *Pluchea indica*, which was collected in July 2015 from Shankou Mangrove Nature Reserve in Guangxi Province, China. The fungus was obtained using a standard protocol for the isolation. Fungal identification was

carried out using a molecular biological protocol by DNA amplification and sequencing of the ITS region, as described previously.²⁰ A BLAST search result showed that the sequence was the most similar (99%) to the sequence of a *Didymella* sp. (compared to HM012812.1). The sequence data obtained from the fungal strain have been deposited at GenBank with accession no. MG571637. A voucher strain was deposited in the Guangdong Microbial Culture Center under patent depository number GDMCC 60100.

Fermentation, Extraction, and Isolation. The fungus was cultured on autoclaved rice solid-substrate medium in 60 Erlenmeyer flasks (1 L), each containing 50 g of rice, 1.5 g of artificial sea salts, and 50 mL of 0.3% saline, at room temperature under static conditions and daylight for 30 days. Following incubation, the mycelia and solid rice medium were extracted three times with EtOAc. The extract was evaporated under reduced pressure to yield 60 g of residue. The residue was subjected to silica gel CC, eluting with a gradient of petroleum ether/EtOAc from 1:0 to 0:1, to obtain 36 fractions. Fraction 6 (120 mg) was subjected to silica gel CC ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ v/v, 98:2) to yield compound **1** (6.8 mg). Fraction 12 (90 mg) was applied to Sephadex LH-20 CC ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ v/v, 1:1) to give compounds **2** (5.2 mg) and **4** (3.4 mg). Fraction 16 (68 mg) was purified by Sephadex LH-20 CC ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ v/v, 1:1) and silica gel CC ($\text{CH}_2\text{Cl}_2/\text{MeOH}$ v/v, 88:12) to afford **3** (4.6 mg). Fraction 20 (62 mg) was chromatographed on Sephadex LH-20 CC (100% MeOH) to obtain compound **5** (1.8 mg).

Ascomylactam A (1): white hydrate crystals; mp 241–243 °C; $[\alpha]_D^{25} +39$ (c 0.35, MeOH); UV (MeOH) λ_{max} (log ϵ) 265 (3.98) nm; IR (KBr) ν_{max} 3469, 3322, 2937, 1661, 1509, 1411, 1368, 1298, 1223, 1110, 1051 cm^{-1} ; ^1H and ^{13}C NMR (500 MHz, CDCl_3) data, Table 1; HRESIMS m/z 558.3222 $[\text{M} - \text{H}]^-$ (calcd for $\text{C}_{35}\text{H}_{44}\text{NO}_5$, 558.3225).

Ascomylactam B (2): white methanolate crystals; mp 250–252 °C; $[\alpha]_D^{25} +98$ (c 0.72, MeOH); UV (MeOH) λ_{max} (log ϵ) 223 (4.36) nm; IR (KBr) ν_{max} 3289, 2937, 1699, 1605, 1505, 1373, 1236, 1013 cm^{-1} ; ^1H and ^{13}C NMR (500 MHz, CDCl_3) data, Table 1; HRESIMS m/z 526.2960 $[\text{M} - \text{H}]^-$ (calcd for $\text{C}_{34}\text{H}_{40}\text{NO}_4$, 526.2962).

Ascomylactam C (3): white solid; $[\alpha]_D^{25} +69$ (c 0.16, MeOH); UV (MeOH) λ_{max} (log ϵ) 238 (4.56) nm; ECD (c 1.2×10^{-3} M, MeOH) λ_{max} ($\Delta\epsilon$) 225 (−17.20), 250 (−8.30), 275 (1.50), 315 (−3.50) nm; IR (KBr) ν_{max} 3495, 3281, 2921, 1740, 1715, 1508, 1450, 1372, 1281, 1232, 1118, 922, 759 cm^{-1} ; ^1H and ^{13}C NMR (500 MHz, $\text{MeOH}-d_4$) data, Table 1; HRESIMS m/z 542.2898 $[\text{M} - \text{H}]^-$ (calcd for $\text{C}_{34}\text{H}_{40}\text{NO}_5$, 542.2906).

X-ray Crystallographic Analysis of Compounds 1, 2, and 5. Single crystals of **1**, **2**, and **5** were obtained from MeOH solution. The data were obtained on an Agilent Xcalibur Nova single-crystal diffractometer using Cu $K\alpha$ radiation ($\lambda = 1.5418$ Å). The structures were solved by direct methods using SHELXS-97. Refinements were performed with SHELXS-97 using full-matrix least-squares, with anisotropic displacement parameters for all the non-hydrogen atoms. Hydrogen atoms on oxygen were located in difference Fourier maps and refined isotropically. Hydrogen atoms bonded to carbons were placed in calculated positions and refined isotropically with a riding method. The CCDC data can be obtained free of charge from the Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK (fax: 44-(0)1223-336033; e-mail: deposit@ccdc.cam.ac.uk).

Crystal data of 1: $\text{C}_{35}\text{H}_{45}\text{NO}_5 \cdot \text{H}_2\text{O}$, $M_r = 577.3$, monoclinic, $\alpha = 11.6923(10)$ Å, $b = 7.0558(10)$ Å, $c = 19.0143(2)$ Å, $\alpha = 90.00^\circ$, $\beta = 100.31^\circ$, $\gamma = 90^\circ$, $V = 1543.28(3)$ Å³, space group $P2_1$, $Z = 1$, $T = 150$ K, $D_{\text{calcd}} = 1.204$ g/cm³, $\mu = 0.631$ mm^{−1}, and $F(000) = 604.0$. Crystal dimensions: $0.40 \times 0.30 \times 0.20$ mm³. Independent reflections: 5447 ($R_{\text{int}} = 0.0365$). The final R_1 values were 0.0378, $wR_2 = 0.0989$ ($I \geq 2\sigma(I)$). The goodness of fit on F^2 was 1.056. The Flack parameter value was −0.13(9). CCDC number: 1515168.

Crystal data of 2: $\text{C}_{34}\text{H}_{41}\text{NO}_4 \cdot \text{CH}_3\text{OH}$, $M_r = 545.3$, monoclinic, $\alpha = 20.9402(10)$ Å, $b = 7.0781(10)$ Å, $c = 20.5614(2)$ Å, $\alpha = 90.00^\circ$, $\beta = 91.39(10)^\circ$, $\gamma = 90^\circ$, $V = 3046.65(5)$ Å³, space group C_2 , $Z = 1$, $T = 150$ K, $D_{\text{calcd}} = 1.173$ g/cm³, $\mu = 0.722$ mm^{−1}, and $F(000) = 1135.0$. Crystal dimensions: $0.42 \times 0.38 \times 0.34$ mm³. Independent reflections:

5621 ($R_{\text{int}} = 0.0487$). The final R_1 values were 0.0449, $wR_2 = 0.1204$ ($I \geq 2\sigma(I)$). The goodness of fit on F^2 was 1.079. The Flack parameter value was −0.04(12). CCDC number: 1818005.

Crystal data of 5: $\text{C}_{34}\text{H}_{41}\text{NO}_4$, $M_r = 527.3$, orthorhombic, $\alpha = 12.1967(2)$ Å, $b = 16.4091(3)$ Å, $c = 47.13(10)$ Å, $\alpha = 90.00^\circ$, $\beta = 90^\circ$, $\gamma = 90^\circ$, $V = 9433.90(3)$ Å³, space group $P2_12_12_1$, $Z = 3$, $T = 293$ K, $D_{\text{calcd}} = 1.430$ g/cm³, $\mu = 1.137$ mm^{−1}, and $F(000) = 4158.0$. Crystal dimensions: $0.42 \times 0.38 \times 0.34$ mm³. Independent reflections: 18 890 ($R_{\text{int}} = 0.0745$). The final R_1 values were 0.0843, $wR_2 = 0.2307$ ($I \geq 2\sigma(I)$). The goodness of fit on F^2 was 1.050. The Flack parameter value was −0.28(9). CCDC number: 1818006.

MeOH Incubation Experiments with Compound 2. A solution of **2** (2.0 mg) in MeOH (2 mL) was stirred at 50 °C for 72 h. But, no change was observed. Then, a catalytic amount of silica gel was added to the solution, which was stirred at 50 °C for 48 h. However, no additional products were detected. Finally, a 1.0 M aqueous HCl solution was added to the solution, which was stirred for 24 h at 50 °C. Still no compound **1** was detected.

Computational Methods. Molecular Merck force field (MMFF) and DFT/TD-DFT calculations were carried out with Spartan'14 software (Wavefunction Inc.) and the Gaussian 09 program, respectively. The conformers with a Boltzmann distribution over 1% were chosen for ECD calculations, and the conformers were generated and optimized using DFT calculations at the B3LYP/6-31G(d) level in MeOH using the CPCM polarizable conductor calculation model. The theoretical calculations of the ECD spectra were conducted in MeOH using time-dependent density functional theory (TD-DFT) at the B3LYP/3-21G level for compounds **3** and **4** and at the B3LYP/sto-3G* level for compound **5**. The ECD spectra were generated using the program SpecDis 1.6 (University of Würzburg) and OriginPro 8.5 (OriginLab, Ltd.) from dipole-length rotational strengths by applying Gaussian band shapes with sigma = 0.30 eV.

Cytotoxicity Assay. Standard MTT assays employing MDA-MB-435, MDA-MB-231, SNB19, HCT116, NCI-H460, and PC-3 cell lines were carried out as described previously.²⁰ The detailed process is described in the Supporting Information.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jnatprod.8b00918.

X-ray data for **1** (CIF)

X-ray data for **2** (CIF)

X-ray data for **5** (CIF)

Selected 1D and 2D NMR spectra of compounds **1**–**5**, conformers, atomic coordinates, energies of the computed structures of **3**–**5**, and cytotoxicity assay (PDF)

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Notes

The authors declare no competing financial interest.

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